



BIRSA INSTITUTE OF TECHNOLOGY SINDRI, DHANBAD — JHARKHAND
DEPARTMENT OF CIVIL ENGINEERING
Estd. 1949 · Jharkhand University of Technology



Birsa Institute of Technology Sindri, Dhanbad · Jharkhand · Estd. 1949

Department of Civil Engineering · Birsa Institute of Technology Sindri

AVLOKAN

Association of Civil Engineers — ACE
Annual Technical Fest · 2026

HACKATHON — DAY I

Official Problem Statement Booklet

12 HOURS
9:00 AM – 9:00 PM
Continuous Sprint

44 CHALLENGES
Across 5 Tracks
Inter-College

TEAMS
3 – 4 Members
Per Team

PRIZES
■ 36,000+ Pool
+ Trophies

avlokan.acebits.in | acebitsindri@gmail.com | Dept. of Civil Engineering, B.I.T. Sindri — 828 123

avlokan.acebits.in

acebitsindri@gmail.com | Dept. of Civil Engineering, B.I.T. Sindri — 828 123

© 2026 Association of Civil Engineers (ACE), B.I.T. Sindri. All Rights Reserved.

For Academic and Competitive Use Only



ABOUT AVLOKAN 2026

AVLOKAN is the annual technical festival of the **Association of Civil Engineers (ACE)**, Department of Civil Engineering, Birsa Institute of Technology (B.I.T.) Sindri. Conducted over three days, AVLOKAN brings together engineering students, faculty, and industry professionals from across the country to collaborate, compete, and innovate. The festival serves as a platform for students to apply classroom theory to real-world problems — with a particular emphasis on civil infrastructure, sustainable development, and digital engineering.

Day	Event	Description
I	HACKATHON <i>12-Hour Sprint</i>	Teams from different colleges choose a problem statement and develop a working prototype — hardware or software — within 12 continuous hours. Emphasis is on practical, deployable, and civil-engineering-relevant solutions. Expert panel evaluates live demonstrations at the end of the day.
II	TECHNICAL EVENTS <i>Paper Presentation, CAD, Quiz</i>	Day 2 features individual and team-based competitions: Technical Paper Presentation, AutoCAD Drafting Contest, and Civil Engineering Quiz. Participants demonstrate domain knowledge, research acumen, and presentation skills before a panel of faculty and industry judges.
III	CULTURAL & VALEDICTORY <i>Fest, Awards & Closing</i>	The concluding day celebrates achievements with cultural performances, guest lectures from industry leaders, prize distribution for all events, and the formal valedictory ceremony. Outstanding Hackathon projects are showcased in a public exhibition.



HACKATHON — RULES & REGULATIONS

1	Eligibility & Registration
1.1	The Hackathon is open to all undergraduate and postgraduate engineering and technology students currently enrolled in a recognised university or institute.
1.2	Participants must register as a team. Each team shall comprise a minimum of 3 and a maximum of 4 members . All team members must be from the same institution.
1.3	Each institution may field a maximum of 5 teams in a single track. A team may register for only one track.
1.4	One member of each team must be designated as the Team Leader , who shall serve as the primary point of contact with the organisers.
1.5	Valid college ID cards are mandatory for all participants and must be presented at the registration desk on Day I. No exceptions shall be made.

2	Problem Statement Selection
2.1	Teams must choose exactly one problem statement from this booklet during online pre-registration, at least 72 hours before the event .
2.2	Problem statement selection is on a first-come, first-served basis. A single problem may be attempted by a maximum of 3 teams across the entire event.
2.3	Teams are strongly encouraged to conduct preliminary research on their chosen problem before the event. Pre-event preparation of concept sketches, algorithm outlines, or system architecture is permitted and encouraged.
2.4	Change of problem statement after registration is not permitted under any circumstance.

3. Event Schedule — Day I (Hackathon)

Time	Activity
08:00 – 09:00	Participant Registration, ID Verification & Kit Distribution
09:00 – 09:30	Inauguration Ceremony, Welcome Address & Track Briefing by Coordinators
09:30 – 09:45	Problem Statement Confirmation & Workstation Allocation
09:45 – 10:00	Final Rules Briefing by the Jury Panel
10:00	◆ HACKATHON BEGINS — 12-Hour Development Sprint ◆
13:00 – 14:00	Lunch Break (Teams must leave a representative at workstation)
16:00 – 16:30	Mid-Point Advisory Review by Faculty Mentors (no marks; guidance only)
19:00 – 19:30	Dinner Break (Teams may not leave premises unattended)



22:00 – 22:15	■ FINAL SUBMISSION DEADLINE — Hard close; no extensions granted
22:15 – 22:45	Repository / Hardware lock-in & Jury Preparation
22:45 – 23:45	Final Demonstrations & Jury Evaluation (Q&A; per team)
23:45 – 00:00	Score Compilation & Day I Results Announcement

4	Deliverables & Submission Requirements
4.1	Teams must submit a working prototype — hardware, software, or a combination of both — that directly addresses their chosen problem statement.
4.2	A Project Report (maximum 8 pages, A4, PDF format) must be submitted digitally by 22:00 hrs. It must include: Problem Definition, Proposed Solution, System Architecture / Design Schematic, Tools & Technologies Used, and Future Scope.
4.3	A 5-minute verbal presentation followed by a 5-minute Q&A with the jury is mandatory for every team. Teams must demonstrate the prototype live during evaluation.
4.4	For software-based solutions: all source code must be committed to a private GitHub repository and the link shared with organisers by the submission deadline.
4.5	For hardware-based solutions: a working prototype or a detailed labelled mock-up with clear functionality demonstration is required. CAD drawings or circuit schematics must be included in the report.

5. Judging Criteria & Scoring Rubric

S.N o.	Criterion	Ma x.	Evaluation Focus
5.1	Problem Understanding & Solution Relevance	20	Does the solution directly and correctly address the stated problem?
5.2	Innovation & Technical Depth	20	Novelty of approach; engineering complexity; use of advanced methods.
5.3	Functionality & Working Prototype Quality	25	Does the prototype / software work as intended? Is it stable and functional?
5.4	Civil Engineering Applicability & Real-World Impact	15	Practical relevance to civil / infrastructure domain; scalability potential.
5.5	Presentation, Report Quality & Team Coordination	10	Clarity of explanation, quality of report, Q&A; handling.
5.6	Sustainable & Responsible Design	10	Sustainability, safety, cost-effectiveness, and ethical considerations.
	TOTAL	100	

6	Resources, Tools & Infrastructure
---	-----------------------------------



6.1	Each team will be allocated a dedicated workstation comprising a table, chairs, and a power outlet. Participants must bring their own laptops, chargers, and any personal tools.
6.2	Stable Wi-Fi connectivity will be provided. Teams requiring basic hardware components (Arduino, sensors, resistors) may request from the resource desk — subject to availability and on a best-effort basis.
6.3	Use of open-source software, libraries, APIs, and publicly available datasets is permitted. Pre-written frameworks are allowed; however, at least 60% of the solution must be developed during the event.
6.4	Faculty mentors from the Department of Civil Engineering will be available for technical guidance at scheduled intervals. Mentors are advisory only and will not assist in building solutions.
6.5	Use of Generative AI tools (ChatGPT, Copilot, etc.) is permitted as an assistive tool only. AI-generated content must be clearly identified in the final report.

7 Code of Conduct & Disciplinary Policy

7.1	All participants are expected to maintain professional conduct. Any form of plagiarism, copying of another team's work, or submission of pre-built projects will result in immediate disqualification.
7.2	Participants must not tamper with or interfere with another team's workstation, equipment, or code.
7.3	The premises must be kept clean and tidy. Damage to institute property will result in disqualification and may attract additional penalty at the discretion of the organising committee.
7.4	Consumption of alcohol, tobacco, or any prohibited substance on campus is strictly forbidden and will result in immediate expulsion from the event.
7.5	The decision of the Jury Panel and the Organising Committee shall be final and binding.
7.6	Exit and re-entry are permitted for emergencies only, with prior written permission from the Event Coordinator. Exit without permission will be treated as voluntary withdrawal.

8. Awards & Recognition

Position	Award Title	Prize	Additional Perks
WINNER	AVLOKAN Grand Innovator Award	■ 15,000 + Trophy	Certificate, Fast-track Industry Referral
1st Runner-Up	Excellence in Engineering Award	■ 10,000 + Trophy	Certificate, Mentorship Session
2nd Runner-Up	Distinguished Innovation Award	■ 5,000 + Trophy	Certificate of Merit
Best Civil Application	Civil Engineering Impact Award	■ 3,000 + Plaque	Special Jury Commendation
Best Sustainability	Green Engineering Award	■ 3,000 + Plaque	Special Jury Commendation



All
Participants

Participation Certificate

—

Issued by Dept. of Civil Engineering, B.I.T. Sindri

Note: Prize money is subject to sponsorship confirmation. The organising committee reserves the right to revise the prize structure with prior notice.



PROBLEM STATEMENTS — ALL TRACKS

The following 44 problem statements are classified into five thematic tracks. Teams must select one problem from within their registered track. Each problem is designed to challenge teams to apply engineering principles to practical, real-world scenarios with measurable outcomes that reflect real civil and infrastructure engineering challenges.

Code	Track Title	No.	Solution Type
TRK-A	Smart Infrastructure & Transportation Engineering	8	Hardware / Software / Simulation
TRK-B	Sustainable Energy & Environmental Engineering	8	Hardware / Software / Prototype
TRK-C	Digital Construction & Structural Health Monitoring	8	Hardware / Software / Simulation
TRK-D	Smart Cities & Urban Systems	10	Software / App / Platform
TRK-E	Open Innovation	10	Hardware / Software / Any



TRK-A — Smart Infrastructure & Transportation Engineering

Hardware / Software
/ Simulation

Focus Area: Urban transportation systems, road & bridge technology, vehicular engineering, traffic management, and mobility optimisation using AI and IoT.

01

Innovative Cooling Solution for Parked Vehicles in Hot Climates

Hardware /
Software

Problem Statement:

Design a sustainable, green-energy-based solution that actively maintains the interior temperature of a parked vehicle at approximately 25°C, regardless of external heat conditions. The solution must not rely on sun shades, blinds, tinted windows, or basic solar fans.

Constraints & Considerations:

1. Must operate effectively in direct sunlight and high ambient temperatures.
2. Should be compliant with regulatory norms (e.g., no tinted windows or external blinds).
3. Should utilise renewable energy sources (e.g., solar, thermoelectric, passive cooling).
4. Should be compact, vehicle-compatible, and ideally retrofit-friendly.

02

Passenger Comfort Modelling in an Automotive Suspension System

Simulation /
Software

Problem Statement:

Establish and validate a mathematical model that accurately represents passenger comfort parameters within a defined suspension system under real-time driving conditions. The model should be supported by physical testing to ensure correlation between theoretical predictions and actual ride experiences.

05

Standardised Battery Swapping Station for All Types of Electric Vehicles

Hardware /
Software

Problem Statement:

Design a standardised, user-friendly battery swapping station that supports all types of EVs with a universal battery concept. The solution should integrate with existing fuel station infrastructure, include robust battery management systems, and offer quick swapping with digital payment systems.

07

Framework for On-Road Dynamic Automobile Pantograph Mechanism

Hardware /
Simulation

Problem Statement:

Develop a framework for powering electric vehicles directly from external overhead power lines by designing a vertical roller pantograph mounted on the vehicle — similar to train pantographs adapted for road vehicles. A working prototype should demonstrate proof of concept.

**08****Open Road Frequency — The Future of Car-to-Car Communication***Hardware /
Software***Problem Statement:**

Design and prototype a system that enables real-time, localised communication between vehicles, inspired by aviation-style shared frequencies. The solution should define a communication protocol, ensure a distraction-free interface, prioritise messages by urgency, and address security, privacy, and V2V/V2X interoperability.

09**CAE Analysis of Automotive Systems with ECU and Virtual Operating Conditions***Simulation /
Software***Problem Statement:**

Develop a solution integrating system CAD models with ECU software to perform dynamic CAE/CFD analysis under virtual operating conditions, with a focus on thermal analysis for electric and hybrid vehicle designs.

31**High-Fidelity Prediction of Stiffness in Stamped Panels Using AI/ML***Software /
Simulation***Problem Statement:**

Develop a scalable AI/ML solution that accurately predicts the stiffness and deformation behaviour of stamped components based on geometric shape and applied load, replacing time-intensive CAE simulations with a fast, high-fidelity predictive model.

39**AI-Driven Urban Mobility Optimisation for Sustainable Smart Cities***Software / Platform***Problem Statement:**

Develop an intelligent real-time urban mobility management solution leveraging AI, machine learning, and IoT to analyse traffic flow data, identify congestion hotspots, and dynamically optimise signal timings and routing. The system should integrate GPS, traffic cameras, and public transport networks into a unified traffic intelligence dashboard.

**TRK-B — Sustainable Energy & Environmental Engineering***Hardware / Software
/ Prototype*

Focus Area: Renewable energy systems, EV technology, energy storage, wind & solar infrastructure, sustainability assessment frameworks, and environmental impact analysis.

04**Enhancing Energy Density of EV Batteries to Overcome Range Anxiety***Hardware /
Research
Prototype***Problem Statement:**

Explore and develop innovative approaches to increase the energy density of EV batteries, extending driving range without compromising safety or manufacturability. The solution should consider advancements in battery chemistry, cell architecture, thermal management, and materials science.

06**Fire Suppression Device for Electric Vehicle Battery Safety***Hardware
Prototype***Problem Statement:**

Design a compact, installable device for EVs that detects abnormal temperature rise in battery packs and automatically releases stored CO₂ to suppress fire at the onset. The system must include temperature sensors, a CO₂ storage and release mechanism, and operate autonomously.

12**Sustainability Rating and Grading Framework for Construction Products***Software /
Framework***Problem Statement:**

Develop a structured process for deriving sustainability ratings for construction products at both part and system levels using life cycle assessment methodologies, considering material composition, design attributes, manufacturing processes, and intended use.

14**AI-Based Alarm Tracking System to Reduce Lost Production in Wind Turbines***Software / AI Tool***Problem Statement:**

Develop an AI-based tool that automates tracking and resolution of wind turbine alarms. The system should apply DMAIC root cause analysis, identify responsible stakeholders, trigger notifications, provide historical analytics, and recommend actionable solutions.

15**GenAI-Based Fault Detection and Resolution in Solar Farms***Software / AI Tool***Problem Statement:**

Build a GenAI-powered solution that accurately detects faults in solar panels using real-time monitoring, automated data collection, and advanced image analysis, supporting predictive maintenance and integration with existing solar farm management tools.

**38****Super-Capacitor or Hybrid Energy Storage System for Lighting Applications***Hardware /
Prototype***Problem Statement:**

Develop a supercapacitor-based or hybrid energy storage system combining rapid charge/discharge capabilities of supercapacitors with high energy density of batteries. This approach addresses limitations of traditional battery-only systems, delivering instant power delivery, extended operational life, and improved energy retention for lighting applications.

40**Model-Based Design Concept for a System with Two Isolated Vibrating Systems***Simulation /
Software***Problem Statement:**

Develop a model-based multibody dynamic simulation framework capable of predicting the vibration behaviour of two or more isolated systems within a shared enclosure, using line body techniques to simplify computational requirements and enable early-stage structural risk assessment.

44**AI-Based Cost-Effective Agile Mobile Blade Balancing for Large Wind Turbines***Hardware / AI***Problem Statement:**

Develop an AI-based, cost-effective, agile, and mobile blade balancing solution for wind turbine blades on 10 MW+ machines, where manual centre-of-gravity deviations currently cause turbine instability, premature bearing damage, and reduced turbine lifespan.

**TRK-C — Digital Construction & Structural Health Monitoring***Hardware / Software
/ Simulation*

Focus Area: Non-destructive testing, construction equipment automation, structural monitoring, robotics in construction, AI-based inspection, and digital quality assurance.

03**AI-Driven Non-Destructive Inspection of Automotive Composite Materials***Hardware /
Software***Problem Statement:**

Develop an AI-enabled non-destructive evaluation system using ultrasound, X-ray, and thermography to detect micro-damages in composite materials without disassembly. The system must incorporate FMEA, lifecycle assessment, and residual life estimation while adhering to ISO/ASTM standards.

10**Autonomous Drones: Edge AI-Enabled Surveillance Platform for Construction Sites***Hardware /
Software***Problem Statement:**

Design and build an autonomous drone platform with Edge AI capabilities for real-time surveillance and threat detection over large construction sites. The system must demonstrate onboard video processing, object detection, and decision-making without cloud dependency.

11**Predictive Maintenance and Safety in Construction Equipment Operations***Software / App***Problem Statement:**

Design an AI-powered web and mobile application that transforms construction equipment operations by integrating operator health monitoring and predictive maintenance. The system should use wearable data to assess stress and fatigue, and apply machine learning to predict equipment failures.

21**Search Bot for CAD Drawing and Model Retrieval in Engineering Projects***Software / AI Tool***Problem Statement:**

Develop a search bot capable of intelligently retrieving CAD drawings and model contents from a large engineering database using keywords, numerical specifications, shapes, and model component queries. A query like 'tank of xx capacity' should yield matching drawings instantly.

24**AI and Computer Vision-Based Non-Destructive Testing for Industrial Materials***Software / AI
Model***Problem Statement:**

Develop an AI model using computer vision to automatically detect and annotate defects in NDT images, trained to identify crack locations, depths, and types. The system should handle industrial-grade stainless steel or aluminium and use an 80/20 dataset split.

**32****Automated Validation of Flow Instrument Installation in 3D Piping Models***Software /
Simulation***Problem Statement:**

Develop an intelligent automated solution that analyses 3D piping models to validate flow instrument installations, measuring upstream/downstream straight pipe lengths, identifying fluid flow direction, and verifying rotameter orientation using AI or rule-based logic.

36**Flexible Robotic Arm for Safe and Adaptive Industrial Operations***Hardware /
Robotics***Problem Statement:**

Design a flexible robotic arm system for manufacturing setups, supporting adaptive gripping for objects of various sizes, reserve power for safe release during power failures, automatic homing, minimised cabling, and improved modularity for ease of installation.

37**Self-Driving Pathological Lab: Robotic Automation for Medical Testing***Hardware /
Robotics***Problem Statement:**

Design and deploy a mobile robotic automation system for pathology labs featuring a Cobot mounted on an AMR, equipped with a precision vision system, modular grippers, and LIMS integration, meeting high standards of accuracy and compliance for medical environments.

**TRK-D — Smart Cities & Urban Systems**Software / App /
Platform

Focus Area: Urban digital infrastructure, smart utilities, healthcare integration, workforce management, communication networks, and data-centric urban services.

19**Chambered Full-Body Walkthrough Diagnostic Tool for Non-Invasive Health Screening**Hardware /
Software**Problem Statement:**

Design a multidimensional, non-invasive diagnostic walkthrough chamber that enables a complete health check-up while the individual simply walks through it. The system should passively monitor key physiological systems and automatically share results via a mobile application.

20**Smart Toilet for Automated Urine and Health Analysis**Hardware /
Software**Problem Statement:**

Develop a smart toilet system with integrated sensors that automatically collect and analyse biological samples. The system should include hygienic sample collection mechanisms, cloud-based diagnostic data storage, and seamless sensor integration without disrupting user comfort.

25**Autonomous Networks for 5G Network Digital Twins**Software /
Simulation**Problem Statement:**

Develop a digital twin-based solution for autonomous 5G network management aligned with xNF deployment standards, supporting interoperability, efficient performance monitoring, and predictive maintenance, adaptable to emerging 6G use cases.

26**Scalable AI-Driven Workforce Digital Twin for Manufacturing Sites**

Software / Platform

Problem Statement:

Propose a Workforce Digital Twin platform that models and simulates the workforce layer of a manufacturing plant. The system should capture real-time IoT and wearable signals to predict optimal workforce deployment and recommend dynamic worker reallocation across multiple sites.

27**Automation in Drawing and Datasheet Conversions Using AI and Gen-AI**

Software / AI Tool

Problem Statement:

Develop an AI/GenAI-based cloud solution that reads and converts engineering drawings and datasheets from any input format — including scanned images and handwritten documents — into native formats compatible with 1D and 2D engineering applications within 120 seconds.

**28****AI-Driven BOM Converter***Software / AI Tool***Problem Statement:**

Design and develop an AI-powered BOM Converter that automates the transformation of engineering Bill of Materials (eBOM) into manufacturing BOM (mBOM). The solution should intelligently restructure the BOM, incorporate manufacturing details, and optimise it for production workflows.

29**AI-Powered Video Localisation Across Multiple Languages***Software / AI Tool***Problem Statement:**

Design an AI-driven solution capable of processing high-quality, large-sized video files and localising them into multiple languages, automating transcription, translation, and audio dubbing while maintaining contextual accuracy and synchronisation.

34**Invisible Watchdogs: Wi-Fi Sensing for Home Monitoring and Surveillance***Hardware /
Software***Problem Statement:**

Design a Wi-Fi Sensing-based solution for indoor surveillance that leverages signal disruptions caused by movement to detect presence and activity — enabling intrusion detection, smart lighting, elder care monitoring, and foot traffic analysis without compromising privacy.

35**LiveSpeak: AI-Based Live Event Captioning via Speech Recognition***Software /
Embedded***Problem Statement:**

Develop a real-time speech-to-text solution for live events ensuring high accuracy and low latency, evaluating cloud-based services and exploring Edge AI approaches for on-device processing using AI-enabled SoCs, deployable across various live event scenarios.

43**Bringing Healthcare to Everyone: Real-Time Monitoring and AI Solutions***Hardware /
Software***Problem Statement:**

Design an AI-powered healthcare solution enabling real-time patient monitoring and prioritisation. The system should include a wearable device for vital sign tracking, a remote clinician dashboard, and an AI triage model. The solution must be cost-effective, scalable, and suitable for rural deployment.

**TRK-E — Open Innovation**Hardware / Software
/ Any

Focus Area: Cross-domain technology challenges spanning AI/ML, cybersecurity, automation, embedded systems, and general-purpose engineering software tools.

13**Intelligent Rubrics-Based Evaluator for Flowcharts, Algorithms, and Pseudocode**

Software / AI Tool

Problem Statement:

Develop an intelligent evaluation system capable of assessing flowcharts, algorithms, and pseudocode against predefined rubrics. The system should parse .docx, .pptx, .pdf, .png, and .jpg files, analyse submissions, and generate detailed feedback with LMS integration support.

16**Automated Electrical Wiring Diagram Generation Using GenAI**

Software / AI Tool

Problem Statement:

Develop a GenAI-powered solution that automatically generates electrical wiring diagrams from electrical panel drawings using image recognition and schematic generation techniques to streamline documentation and reduce errors.

17**Automating Material Selection for Design Projects Using GenAI**

Software / AI Tool

Problem Statement:

Develop a GenAI-powered assistant that automates the material selection process by analysing application requirements and matching them with suitable materials from global standards. The solution should simulate material behaviour and prioritise sustainable and recyclable options.

18**Automating the Sizing and Selection of COTS Parts Using GenAI**

Software / AI Tool

Problem Statement:

Develop a GenAI-powered solution that automates the sizing and selection of Commercial Off-The-Shelf (COTS) parts based on design parameters. The system should analyse large datasets, simulate part performance, and recommend optimal components considering load, cost, and sustainability.

22**AI-Driven Testing Framework for OTT and Digital TV Ecosystems**

Software / AI Tool

Problem Statement:

Develop an AI-powered automated testing framework for OTT and Digital TV platforms with AI-generated test cases, self-healing test scripts, visual regression testing using deep learning, and predictive test optimisation. CI/CD integration should enable smart test execution.

**23****Middleware to Recommend Enhancements from Non-Agentive to Agentive AI Frameworks***Software / AI Tool***Problem Statement:**

Build a reasoning-driven middleware engine that analyses existing solution documentation and recommends enhancements to convert them into agentive AI systems. The middleware should assess agentive integration feasibility and suggest functional and technical improvements.

30**Agentless Endpoint Protection: Efficient, Reliable, and Cost-Effective***Software /
Cybersecurity***Problem Statement:**

Design a strategic, agentless endpoint protection solution that enhances visibility and threat response capabilities across physical devices, complementing existing agent-based tools by providing coverage for endpoints that are difficult to manage or monitor.

33**AI (Agentive) Enabled Agile Automated Threat Modelling***Software /
Cybersecurity***Problem Statement:**

Design an AI-powered, agentive, and modular threat modelling framework for engineering sector cybersecurity. The solution should adapt to new threat vectors in real time and support use cases such as AI-enabled medical devices, providing automated analysis and risk mitigation.

41**Enhancing Cybersecurity Resilience in the IT Industry Using AI***Software /
Cybersecurity***Problem Statement:**

Develop an AI/ML-powered cybersecurity framework that detects and neutralises emerging threats in real time, reducing detection time by at least 50%. The solution should include automated incident response, seamless integration with existing IT infrastructure, and behavioural analytics-driven training.

42**SmartOps AI — Autonomous Incident Resolution Engine***Software / AI Tool***Problem Statement:**

Build an intelligent, automated solution that detects, analyses, and resolves DevOps incidents with minimal human involvement across hybrid and multi-cloud infrastructures (AWS, Azure, GCP), addressing the reactive and fragmented nature of traditional incident management.



AVLOKAN 2026

Association of Civil Engineers — ACE · B.I.T. Sindri

Department of Civil Engineering

Birsa Institute of Technology Sindri — 828 123, Dhanbad, Jharkhand

acebitsindri@gmail.com

© 2026 Association of Civil Engineers, B.I.T. Sindri. All Rights Reserved.

This document has been prepared by the Organising Committee of AVLOKAN 2026.

For Academic and Competitive Use Only.